

Isobaric Binary Vapor–Liquid Equilibrium of Ethanol + Glycerol and 1-Propanol + Glycerol Systems at 16.0 and 101.3 kPa

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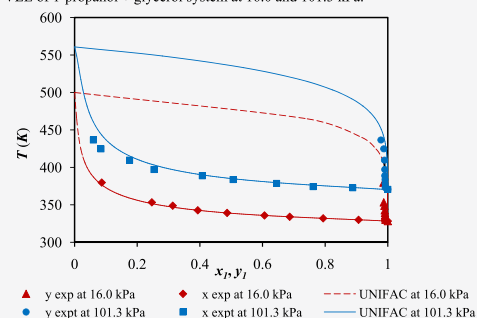
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ABSTRACT: The isobaric binary vapor–liquid equilibrium data of ethanol + glycerol and 1-propanol + glycerol systems were collected experimentally at two pressure levels of 16.0 and 101.3 kPa using modified Glass Othmer–Still. Refractometer index Bausch and Lomb ABBE-type was used to analyze the equilibrium samples of liquid and vapor phase compositions. The reliability of the experimental data determined in this study was conformed to the *L–W* thermodynamic consistency test method proposed by Wisniak (*Ind. Eng. Chem. Res.* 1993, 32(7), 1531–1533). The data were regressed with the Wilson, NRTL, and UNIQUAC equations. The deviations of temperatures and vapor phase compositions were given, and the results indicated that the average absolute deviations in temperatures and in vapor phase compositions were less than 0.5 and 1.1%, respectively, for both systems studied. Furthermore, the performance of predictive UNIFAC equation was found to be better at a pressure of 16 kPa than that of 101.3 kPa.

VLE of 1-propanol + glycerol system at 16.0 and 101.3 kPa.



1. INTRODUCTION

In biodiesel production plants, the processes of transesterification, enzymatic transesterification, hydroesterification, and supercritical transesterification have been implemented in most industries. However, transesterification is the most frequently used process.¹ Transesterification corresponds to a reaction process of oil or fat with an overabundance of alcohol in the presence of a homogeneous or heterogeneous catalyst for the production of biodiesel and glycerol as a product.^{2–4} After cooling to ambient temperature, the glycerol layer settles to the lower and the ester-rich phase (FAME) forms in the upper layer.^{5,6} Glycerol and unreacted alcohol are separated so that alcohol may be recycled for the reaction of transesterification.⁷ Moreover, the glycerol-rich phase will be further processed to produce crude glycerol commonly with a purity of 40–50% containing remaining alcohol component, ester, and other impurities. Therefore, the purification process is needed to achieve the glycerol to meet standard product quality. Commonly, in the transesterification reaction glycerol is known as a commercial by-product that can be used in pharmacy, cosmetics, and food industry.⁸ The vapor–liquid equilibrium (VLE) of the (alcohol + glycerol) data binary system is necessary in an attempt to design and optimize the glycerol production and remove the remaining alcohol.

Many studies have been published related to VLE data of alcohols + glycerol. The isobaric VLE data of binary system alcohol + glycerol have been published by some researchers.^{9–13} Shimoyama et al.¹⁴ measured isothermally the VLE data for system at a temperature of 493–573 K. Moreover, Oliveira et al.⁹ measured and correlated binary VLE for 1-

propanol + glycerol with limited mole fraction ranges of (0.0418 to 0.4410) and ethanol + glycerol with limited mole fraction ranges of (0.0107 to 0.4754) at atmospheric pressure. Wibawa et al.¹⁵ measured the isothermal VLE data for two binary systems of ethanol + glycerol and 2-propanol + glycerol at temperatures of (323.15 and 333.15) K and (333.15 and 343.15) K, respectively. Faggion et al.¹⁶ measured isobaric binary, ternary, and quaternary VLE data (P – T – w_1) at the pressures of (14.0 to 91.2) kPa (ethanol + glycerol), (glycerol + ethanol + NaCl), and (water + ethanol + NaCl + glycerol) systems at 91.2 kPa. Shang et al.¹⁷ measured and correlated ternary VLE data (P – x – y) for methanol + glycerol + methyl palmitate/methyl laurate/methyl myristate systems at 493.0 and 523.0 K. Soujanya et al.¹⁸ measured VLE data of the quaternary system (T – x), methanol + isopropanol + water + glycerol using a Sweitoslawsky-type ebulliometer at 53.33, 66.66, 79.99, and 95.2 kPa. Lemos et al.¹⁹ also determined experimentally vapor–liquid equilibrium data (T – x_1) of (ethanol + glycerol) and (tertbutanol + glycerol) systems.

The most available data regarding the VLE data containing glycerol are T – x and P – x data, and there are only a few available data for T – x – y and P – x – y . For T – x and P – x data,

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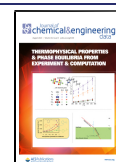


Table 1. Purity and Properties of Pure Components

component	source	CASRN	mass fraction ^a	MW ^b (g mol ⁻¹)	boiling ^c point (K)	density ^d (g cm ⁻³)
ethanol	Merck, Germany	64-17-5	0.999	46.07	351.80 ²⁵	0.78515 ²⁶
1-propanol	Merck, Germany	71-23-8	0.998	60.10	370.93 ²⁵	0.79962 ²⁷
glycerol	Merck, Germany	56-81-5	0.995	92.09	563.00 ²⁸	1.25780 ²⁹

^aPurity from supplier. ^bMW = molecular weight. ^cAt 101.3 kPa. ^dAt 298.15 K and 101.3 kPa.

vapor phase compositions were usually determined by prediction using correlated models. In addition, the quality of data might not be validated using the existing thermodynamic consistency test.

Therefore, in this study the isobaric binary VLE data were determined experimentally using modified Glass Othmer-Still at two pressure levels of 16.0 and 101.3 kPa. The experimental data were correlated with the Wilson,²⁰ nonrandom two-liquid (NRTL),²¹ and universal quasi chemical (UNIQUAC)²² equation. The performance of UNIFAC²³ equation in predicting both systems was studied as well.

2. EXPERIMENTAL SECTION

2.1. Material. Ethanol, 1-propanol, and glycerol used in this experiment were supplied from Merck (Germany) and used without further treatment. Table 1 listed the purity, molecular weight, boiling point, and density of each material.

2.2. Apparatus and Procedures. The isobaric VLE data which were measured using modified Glass Othmer-Still developed in our previous study²⁴ is presented in the schematic diagram in Figure 1. Since the solution containing

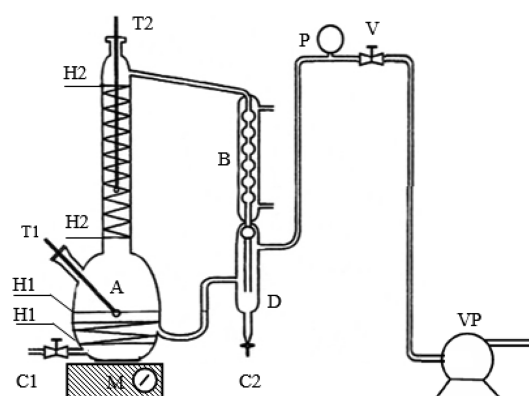


Figure 1. Schematic diagram of modified Glass Othmer-Still. (A) Ebullimeter still, (B) condenser, (C) sample ports, (D) condensate chamber, (H) heater, (T) temperature indicator, (M) magnetic stirrer, (P) pressure indicator, (V) valve control, (VP) vacuum pump.

glycerol is viscous, the modification of the ebullimeter was made by installing the magnetic stirrer in the flat shape lower section of the equilibrium cell in order to make sure the liquid solution in the cell was maintained in a homogeneous condition during equilibrium condition. As shown in the figure, the experimental apparatus consisted of ebullimeter still with a volume of 200 cm³ equipped with a heating control system, condenser, two temperature sensors, autonics RTD Pt-100 Tw-N, and indicators, autonics TZN 4S, vacuum pump with maximum capacity vacuum pressure of 5 Pa, and pressure indicator autonics Ps AN with accuracies of 0.1 K and 0.1 kPa, respectively. The equilibrium temperature was measured by temperature sensor T1, the vapor temperature in the top section of the equilibrium cell was measured by temperature

sensor T2, and the top section temperature was kept about 5 K above the equilibrium temperature by controlling the heater H2 to prevent condensation. The pressure was controlled with a precession of ± 0.2 kPa using a sensitive needle valve and was kept at a constant pressure. The ebullimeter was equipped with two sample ports to collect liquid and vapor phase samples.

The mixture of ethanol + glycerol solution with certain composition was inserted into the equilibrium cell, and all of the valves were closed. In order to remove impurities and air from the ebullimeter and the line package, the vacuum pump was turned on until the constant pressure in the system was achieved. Furthermore, the heating system was turned on until the equilibrium condition was achieved indicated by a constant temperature and pressure of the system. The samples of vapor and liquid phases were collected 3 h after the equilibrium condition was attained from sample ports (C1) and (C2), respectively. Both compositions were analyzed using a Bausch and Lomb refractometer ABBE-type with accuracy in refractive index of ± 0.002 , giving standard uncertainty in the composition of 0.002. For other initial compositions, the experiment was repeated so that the equilibrium composition was determined in the range of 1 to 100% mole. The same procedure was repeated for the 1-propanol + glycerol system.

3. RESULT AND DISCUSSION

The reliability of the experiment was confirmed by measuring vapor pressure of pure ethanol and 1-propanol (see Table 2) at

Table 2. Vapor Pressure of Pure Ethanol and 1-Propanol Measured in This Work^a

ethanol		1- propanol	
P/kPa	T/K	P/kPa	T/K
13.575	308.15	5.205	308.15
17.832	313.15	6.898	313.15
22.869	318.15	9.187	318.15
29.357	323.15	12.137	323.15
37.025	328.15	15.756	328.15
45.928	333.15	19.978	333.15
57.564	338.15	25.495	338.15
72.132	343.15	32.297	343.15

^aStandard uncertainty of measurements are $u(T) = 0.2$ K, $u(P) = 0.2$ kPa, where $u(T)$ is the uncertainty in temperature; and $u(P)$ is the uncertainty in pressure.

the temperatures range of 308.15–348 K, and the data were compared with literature values calculated using the Antoine equation as presented in Figure 2, and the Antoine coefficients are listed in Table 4. The results show that the experiment data agreed well with the literature values giving average absolute deviations (AAD) less than 1%.

The experimental isobaric VLE data (T - x - y) have been measured for two binary systems of ethanol + glycerol and 1-propanol + glycerol at 16.0 and 101.3 kPa, respectively. Table

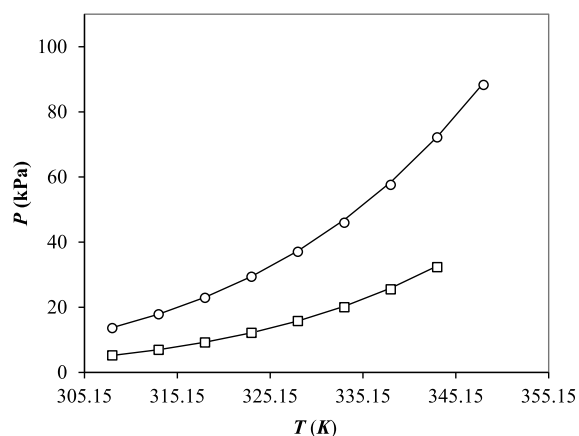


Figure 2. Comparison of pure vapor pressures in experimental data and literature value. Experimental values of (○) ethanol and (□) 1-propanol; the solid line represents the literature values.

3 presents the experimental data obtained in this work. The differences of the boiling temperature values of ethanol and 1-

Table 3. Binary VLE Experimental Data for Ethanol + Glycerol and 1-Propanol + Glycerol Systems at 16.0 and 101.3 kPa^a

ethanol + glycerol			1-propanol + glycerol		
T/K	x_1	y_1	T/K	x_1	y_1
P = 16.0 kPa			P = 16.0 kPa		
310.85	1.0000	1.0000	328.35	1.0000	1.0000
313.45	0.8656	0.9989	330.05	0.9066	0.9924
317.15	0.7432	0.9981	332.05	0.7935	0.9921
319.65	0.6536	0.9939	334.05	0.6867	0.9916
321.55	0.5756	0.9912	335.85	0.6064	0.9911
326.85	0.4571	0.9903	339.25	0.4869	0.9908
331.05	0.3787	0.9899	342.75	0.3935	0.9902
337.05	0.2803	0.9899	348.85	0.3133	0.9891
350.35	0.1662	0.9885	353.35	0.2467	0.9872
366.85	0.0878	0.9871	379.75	0.0867	0.9861
P = 101.3 kPa			P = 101.3 kPa		
351.55	1.0000	1.0000	370.75	1.0000	1.0000
352.35	0.9410	0.9924	372.95	0.8877	0.9921
355.45	0.8377	0.9924	374.35	0.7616	0.9921
356.75	0.7601	0.9924	378.65	0.6447	0.9911
360.05	0.6740	0.9924	383.85	0.5059	0.9911
361.75	0.5055	0.9924	388.85	0.4076	0.9901
369.65	0.4370	0.9876	397.25	0.2536	0.9901
375.85	0.2564	0.9876	409.75	0.1755	0.9891
400.45	0.0680	0.9826	424.85	0.0841	0.9871
425.75	0.0432	0.9777	436.65	0.0598	0.9780

^aThe standard uncertainties are $u(T) = 1.5$ K, $u(x) = 0.002$, $u(y) = 0.002$, and $u(P) = 0.2$ kPa, where $u(T)$, $u(x)$, $u(y)$, and $u(P)$ are the uncertainties in temperature, liquid mole fraction, vapor mole fraction, and pressure, respectively.

propanol listed in Table 1 with the those measured in this work as listed in Table 3 were found to be within the measurement uncertainty.

The experimental data were treated using phase equilibrium criteria, where in equilibrium condition, vapor liquid phases of each component should be isofugacity. For low equilibrium pressure, the equilibrium condition can be written as follows

$$y_i P = x_i \gamma_i P_i^{\text{sat}} \quad (1)$$

where γ_i can be determined from the experimental data, y_i is the mole fractions of component i in the vapor phase, P is the pressure at 16.0 and 101.3 kPa, x_i is the mole fractions of component i in the liquid phase, γ is the activity coefficient of component i in the liquid phase, and P_i^{sat} is saturated vapor pressure of the pure components i at the equilibrium conditions calculated using Antoine equation with the constants obtained from the Thermodynamics Research Center data bank, College Station, TX, U.S.A. The coefficients of Antoine equation for each component are listed in Table 4.

Table 4. Parameter of the Antoine Equation and Activity Coefficients Correlation

component	A	B	C
ethanol ^a	5.33675	1648.220	230.918
1-propanol ^a	4.99991	1512.940	205.807
glycerol ^b	10.6190	4487.040	-140.200

^aReference 25, $\log(P_i^{\text{sat}}) = A - B/(T + C - 273.15)$, where P_i^{sat} is in bar and T is in K. ^bReference 28, $\ln(P_i^{\text{sat}}) = A - B/(T + C)$, where P_i^{sat} is in bar and T is in K.

The L - W test method proposed by Wisniak³⁰ was used to test the thermodynamic consistency of the experimental data.

The thermodynamic consistency data test was evaluated by the following equation

$$L_i = \left[\sum x_i \frac{\Delta s_i^{\text{sat}}}{\Delta s} T_i^{\text{sat}} - T \right] = \left[\frac{g^E}{\Delta s} - \frac{RT}{\Delta s} \omega \right] = w_i \quad (2)$$

where $\Delta s = \sum x_i \Delta s_i^{\text{sat}}$ and $\omega = \sum x_i \ln y_i/x_i$ and the weighted entropy vaporization was evaluated by

$$g^E = RT \sum x_i \left(\ln \frac{y_i}{x_i} + \ln \frac{P}{P_i^{\text{sat}}} \right) \quad (3)$$

To determine the value of $\ln(P/P_i^{\text{sat}})$, the limits are (P_i^{sat}, T) and (P, T_i^{sat}) , where T_i^{sat} is the boiling point at the equilibrium pressure of the system (P), and P_i^{sat} is the vapor pressure of pure component i at the boiling temperature of the solution (T). The experimental data are conformed to the Gibbs–Duhem equation or are thermodynamically consistent if the condition $0.92 < L_i/W_i < 1.08$ is fulfilled. The thermodynamic consistency results are presented in Table 5. As shown in the table, the experimental data were shown to be thermodynamically consistent.

The activity coefficient of each component were regressed with Wilson, NRTL, and UNIQUAC equations. The binary interaction–parameter pairs of each equation were determined by minimizing the following equation

Table 5. Thermodynamic Consistency Test with the L - W Test of Wisniak

system	pressure	$0.92 < L_i/W_i < 1.08$	consistency
ethanol + glycerol	16.0 kPa	0.9965	consistent
	101.3 kPa	1.0257	consistent
1-propanol + glycerol	16.0 kPa	1.0043	consistent
	101.3 kPa	1.0045	consistent

$$\text{OF} = \sum_{i=1}^N \left(\frac{\gamma_1^{\text{exp}} - \gamma_1^{\text{calcd}}}{\gamma_1^{\text{exp}}} \right)^2 + \left(\frac{\gamma_2^{\text{exp}} - \gamma_2^{\text{calcd}}}{\gamma_2^{\text{exp}}} \right)^2 \quad (4)$$

where N represents the number of data points and subscripts exp and calcd refer to experimental and calculated values, respectively.

The binary parameters pairs fitted in this study include AAD between the experimental data and calculated ones in temperatures (AADT) and that in vapor phase compositions (AADy) of each equation and are presented in Table 6. The

Table 6. Binary Interaction Parameters of Wilson, NRTL, and UNIQUAC Equations

	interaction parameter				
equation	Λ_{12}	Λ_{21}	α	AAD ^T (%) ^a	AAD ^y (%) ^b
ethanol + glycerol, $P = 16.0$ kPa					
Wilson	-180.91	588.46		0.1	0.8
NRTL	-10.07	19.20	0.3	0.1	0.8
UNIQUAC	-408.87	624.44		0.1	0.8
UNIFAC				1.0	0.8
ethanol + glycerol, $P = 101.3$ kPa					
Wilson	4300.08	-892.88		0.5	1.1
NRTL	-314.97	1511.84	0.3	0.4	1.1
UNIQUAC	-1466.52	3297.97		0.5	1.1
UNIFAC				8.2	1.1
1-propanol + glycerol, $P = 16.0$ kPa					
Wilson	-1408.48	5215.06		0.1	0.9
NRTL	933.67	-406.94	0.3	0.1	0.9
UNIQUAC	2198.02	-1260.14		0.1	0.9
UNIFAC				1.1	0.8
1-propanol + glycerol, $P = 101.3$ kPa					
Wilson	4099.19	-712.69		0.4	0.8
NRTL	-408.77	1587.85	0.3	0.4	0.9
UNIQUAC	-1501.70	2878.61		0.4	0.8
UNIFAC				7.2	1.0

^a AADT = $\frac{1}{n} \sum_{i=1}^n \left| \frac{T_{\text{cal}} - T_{\text{exp}}}{T_{\text{exp}}} \right| \times 100$ where n is the number of data

point. ^b AADy = $\frac{1}{n} \sum_{i=1}^n \left| \frac{y_{\text{cal}} - y_{\text{exp}}}{y_{\text{exp}}} \right| \times 100$ where n is the number of data point.

table shows that the experimental data were well fitted with three correlated equations giving AADT and AADy less than 0.5 and 1.1%, respectively. The T - x - y data for ethanol + glycerol and 1-propanol + glycerol systems at 16.0 and 101.3 kPa compared with correlated ones are presented in Figures 3, 4, 5, and 6, respectively. The experimental data obtained in this study were well correlated based on Wilson, NRTL, and UNIQUAC equations. Since the different boiling temperatures of ethanol, 1-propanol, and glycerol are wide enough, the vapor phase composition is found to be close to be 1. The T - x - y diagrams for ethanol + glycerol and 1-propanol + glycerol systems at varied pressure of 16.0 and 101.3 kPa are shown in Figures 7 and 8, respectively. In addition, the experimental data for pressure 101.3 kPa for ethanol + glycerol and 1-propanol + glycerol systems were compared with the literature T - x VLE data,⁹ and the figures show that our data agreed well with the literature data with absolute mean deviation in temperatures between literature data and calculated ones using Wilson equation with binary interaction parameters fitted using our

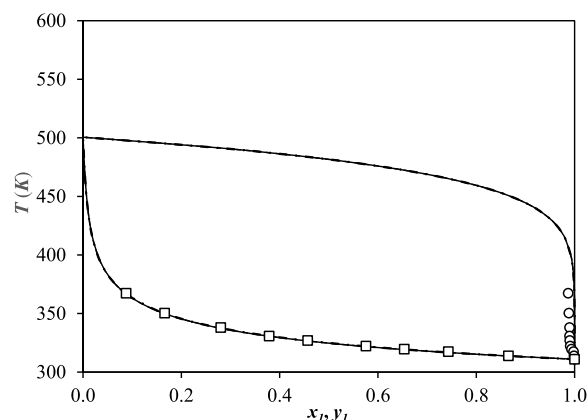


Figure 3. Binary VLE (T - x - y) diagram of ethanol + glycerol system at 16.03 kPa: ($\square$), T - x , for experimental data; (o), T - y for experimental data; (- - -), Wilson equation; (- · -), NRTL equation; (—), UNIQUAC equation.

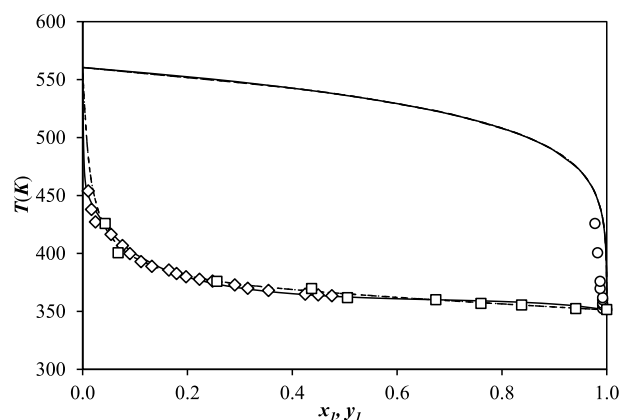


Figure 4. Binary VLE (T - x - y) diagram of ethanol + glycerol system at 101.3 kPa: ($\square$), T - x , for experimental data; (o), T - y for experimental data; ($\diamond$), T - x experimental data of Oliveira et al.;⁹ (- - -), Wilson equation; (- · -), NRTL equation; (—), UNIQUAC equation.

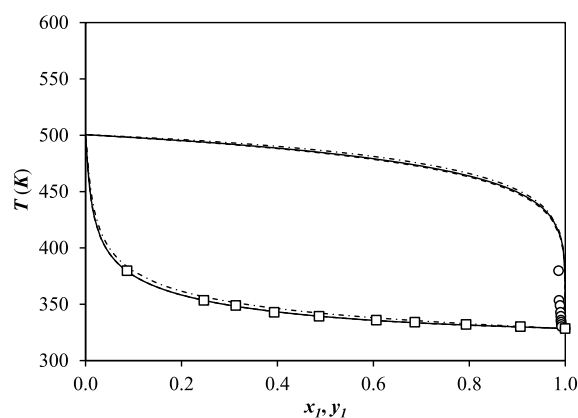


Figure 5. Binary VLE (T - x - y) diagram of 1-propanol + glycerol system at 16.0 kPa: ($\square$), T - x , for experimental data; (o), T - y for experimental data; (- - -), Wilson equation; (- · -), NRTL equation; (—), UNIQUAC equation.

data of 0.7 and 2.2%, respectively. It can be seen that decreasing equilibrium pressure from 101.3 to 16.0 kPa affects the decrease in the equilibrium temperatures significantly.

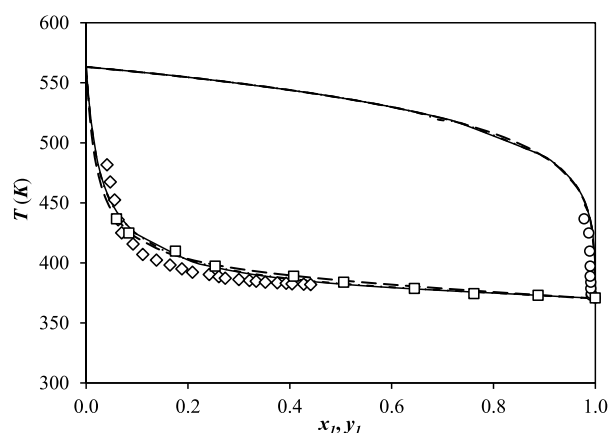


Figure 6. Binary VLE (T - x - y) diagram of 1-propanol + glycerol system at 101.3 kPa: ($\square$), T - x , for experimental data; ($\circ$), T - y for experimental data; ($\diamond$), T - x experimental data of Oliveira et al.;⁹ (---), Wilson equation; (- · -), NRTL equation; (—), UNIQUAC equation.

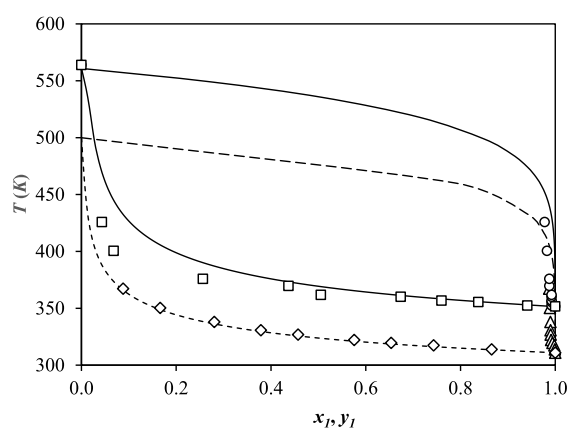


Figure 7. Comparison of experimental data with predictive values using UNIFAC equation for binary VLE of ethanol + glycerol system at 16.0 and 101.3 kPa.

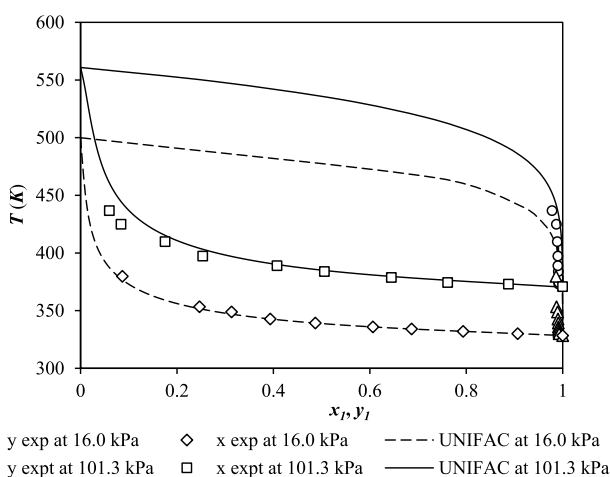


Figure 8. Comparison of experimental data with predictive values using UNIFAC equation for binary VLE of 1-propanol + glycerol system at 16.0 and 101.3 kPa.

The UNIFAC²³ equation was applied to estimate temperature and vapor phase composition based on functional group interaction of each component, that is, the VLE data of the binary system ethanol, 1-propanol, and glycerol, and the functional groups were divided as the interaction parameter of (CH_3 -) with ($-\text{CH}_2$ -) with ($-\text{CH}-$) and with ($-\text{OH}$). The volume R_k and surface area Q_k parameter were obtained from Hansen et al.³¹

For a prediction using the UNIFAC equation, the functional groups were arranged by each of the components listed in Table 7. As shown in Table 6, the prediction results were

Table 7. UNIFAC Group Identification³¹

molecule	group identification		V_k	R_k^{31}	Q_k^{31}
	group	main			
ethanol	CH_3	1	1	0.9011	0.848
	CH_2	1	2	0.6744	0.540
	OH	5	14	1	1.200
1-propanol	CH_3	1	1	0.9011	0.848
	CH_2	1	2	0.6744	0.540
	OH	5	14	1	1.200
glycerol	CH_2	1	2	0.6744	0.540
	CH	1	3	0.4469	0.228
	OH	5	14	3	1.200

compared with the experimental data for ethanol + glycerol and 1 propanol + glycerol systems for pressure of 16.0 kPa giving 1.0% AADT, 0.8% AADy, and 1.1% AADT, 0.8% AADy, respectively at 16.0 kPa. Pressure of 101.3 kPa for ethanol + glycerol and 1-propanol + glycerol systems gave 8.2% AADT, 1.1% AADy, and 7.2% AADT, 1.0% AADy, respectively. The results show that the performance of the UNIFAC equation was better for pressure of 16.0 kPa than that pressure of 101.3 kPa.

4. CONCLUSIONS

In this study, the isobaric binary VLE data for ethanol + glycerol and 1-propanol + glycerol systems at 16.0 and 101.3 kPa were collected experimentally using modified Othmer-Still. The experimental data was conformed to the L - W thermodynamic consistency test method. The experimental data regressed well with the equations of Wilson, NRTL, and UNIQUAC giving AADT and AADy less than 0.5 and 1.1%, respectively. The performance of the predictive UNIFAC equation was found to be better for pressure of 16.0 kPa than that of 101.3 kPa.

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Notes

The authors declare no competing financial interest.

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